Final Testing & Release Document

This document consolidates every test artefact for the **MidnightZK Off-Ramp SDK** at the `v1.0.0` public release, plus the links to the released SDK and the hosted documentation site, so a reviewer can verify "the SDK is operational without any major errors" from one page.

Item	Status
Tagged release	**v1.0.0** — GitHub Release
Hosted developer docs	https://nucastio.github.io/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/
Public GitHub repository	https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments
Recorded demo walkthrough	`docs/media/offramp-demo.mp4`

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1. Internal testing (fresh re-run, 2026-06-13)

A fresh run of the internal harness (`npm run test:internal`) was executed against the v1.0.0 build immediately before tagging the release. Result: **29** / **30 successes (96.7%)**, comfortably above the ≥ 90% acceptance threshold. The single failure is an **intentional negative-path** (the mock-mode adapters inject a ~4–6% adapter-rejection rate to exercise the `RailError` path in `submitPayment`).

Run timestamp: `2026-06-13T10:35:32.919Z` **Total simulated off-ramps:** 30 (10 × cashapp / wise / revolut) **Overall success rate: 96.7%** — PASS (threshold ≥ 90%) **Avg proof-generation latency: 744.6 ms** — PASS (NFR-2 budget ≤ 50 000 ms) **Avg proof-verification latency:** 0.21 ms

Rail	Runs	Successes	Failures	Success rate	Avg prove (ms)	Avg verify (ms)	Avg submit (ms)	Avg attest (ms)
cashapp	10	10	0	100.0%	742.69	0.17	0.27	0.85
wise	10	10	0	100.0%	763.34	0.19	1778.12	0.55
revolut	10	9	1	90.0%	727.76	0.28	0.17	0.46

The full report (per-step latency, methodology, issues-and-fixes) is in `internal-testing-report.md` .

> Reproduce: `npm run test:internal` from a clean checkout. Rewrites `docs/internal-testing-report.md` + `data/testing-report.json` .

2. Cardano Preprod end-to-end

Five real Preprod transactions exercise the **full validator surface** (LOCK / REFUND / RELEASE for both Cash App and Wise paths). All five are confirmed on-chain and linked from `testnet-evidence.md` :

Step	Adapter	Tx hash	Cardanoscan
LOCK #1	Cash App	`f26f023dfc809cb1adad4830bae0025cbe1334fae9811c7a036239eb85fbc6c3`	view
REFUND	sender-signed	`a8c50ba93412a26c5401dc4477ea6307ad56c808c99a174ab8a69c7675c6d0b9`	view
LOCK #2	Wise	`03089ef869daf44c511539c915bc825435c18770071aa923322b43b29dc3b869`	view
LOCK #3	Wise	`b55e48084290f6b88b8fd6489f40e65acc50664fba4873feb1248dffbcb64ac2`	view
RELEASE	operator-signed	`c84c242d6f86dbdac54ded62c92bbdc88b5725722d1691728854e20d62bd3168`	view

- **Validator:** `cardano/escrow/validators/escrow.ak` , Plutus V3, Aiken v1.1.21.
- **Script address (Preprod):** `addr_test1wrvzmkxhfmr9j0u8g6p4cpkevqja4tn8qr88z7l7nc2tqrsxln25g`
- **Datum shape recorded on-chain:** `intent_id` , `payee_commitment` , `amount_commitment` , `adapter_tag` , `sender_pkh` , `operator_pkh` , `deadline` , `vk_hash` , `principal_lovelace` (each LOCK tx in the table above carries a concrete instance).

3. Wise sandbox — live provider integration

The Wise sandbox adapter was driven end-to-end against the public Wise sandbox (`https://api.sandbox.transferwise.tech`) with `RAIL_ADAPTER_MODE=sandbox` . Six raw provider request/response captures are committed under `docs/sandbox-evidence/` , demonstrating real off-ramp flows when credentials are configured.

Step	File	API path	Method	HTTP
1. List profiles	`01-profiles.json`	`/v1/profiles`	GET	200
2. Create quote (USD → USD, 1.50)	`02-quote.json`	`/v3/profiles/{id}/quotes`	POST	200
3. Create recipient	`03-recipient.json`	`/v1/accounts`	POST	200
4. Create transfer	`04-transfer.json`	`/v1/transfers`	POST	200
5. Fund transfer	`05-fund.json`	`/v3/profiles/{id}/transfers/{tid}/payments`	POST	403 (SCA-gated; provider-side)
6. Check transfer status	`06-status.json`	`/v1/transfers/{tid}`	GET	200

Real Wise transfer: `2147582543` , sandbox profile `30539072` , quote `fedc7ae5-c015-4e37-bb71-404104419610` , recipient `702406717` . Status `incoming_payment_waiting` (the documented Wise state for an unfunded transfer; funding completion requires either an OAuth-token profile or Wise SCA approval and is provider-side).

> Cash App uses **Afterpay sandbox** semantics — canonical provider reference: https://www.postman.com/afterpay-1-426879/afterpay-online-apis-v2/folder/zohg5nd/checkouts. Revolut follows the same `RailAdapter` interface and is ready for credentials.

4. Midnight ZK circuit deployment

- **Compact source:** `contract/src/offramp.compact`
- **Compiled artefacts (committed):** `contract/src/managed/`
- **Predicates:** `provePayeeBinding`, `proveAmountBinding`, `proveComplianceFlag`, `proveOffRampSettlement`
- **Contract address:** `a3a72a55eb37317300a6c0718578a8e82040dacce3f139e23e1f52af99f77030`
- **Deploy tx:** `bb81cf19…6e9f2` (block 15768)
- **4 SNARK proofs** deployed across blocks 15774→15784 (per-block detail in `testnet-evidence.md`).

5. Backend API

`tsx backend/api/main.ts` boots a Hono HTTP server that serves the entire off-ramp lifecycle as REST. The OpenAPI 3.0.3 spec is generated by `backend/api/openapi.ts` and exposed live at `/docs` (Swagger UI) and `/api/openapi.json` (machine-readable). All routes are documented in the API reference.

A live evaluation-window deployment is fronted by a TryCloudflare tunnel — see the README for the current URL. The tunnel is ephemeral; reproducing the run requires only `npm run dev` per the quickstart.

6. Issues fixed before release

Issue	Fix	Where
`libsodium-wrappers-sumo` ESM resolution — npm publish layout puts `libsodium-sumo.mjs` in a sibling package; ESM build expects it next to `libsodium-wrappers.mjs`.	One-time shell fixup that copies the file into the expected location.	`scripts/fix-libsodium.sh`
Mock-mode rail rejection wasn't reflected in the harness summary cleanly.	Per-rail success-rate column + dedicated "Failures" section in the regenerated report.	`scripts/internal-test.ts`
Wise sandbox `/payments` SCA-gating.	Documented as expected provider-side behaviour; the SDK's responsibility ends at submitting the funding intent.	`docs/sandbox-evidence/README.md` §"Step 5 note"

7. Acceptance-criteria checklist

Milestone criterion	Result	Evidence
Comprehensive developer docs published (integration, API reference, examples)	✅	Docs site — index / quickstart / integration / api-reference / sdk-reference / examples / architecture
Final testing + release document with all test results, SDK link, docs link	✅	This page
Public GitHub repository with final SDK & docs	✅	https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments
Recorded demo walkthrough video	✅	`docs/media/offramp-demo.mp4`
Tagged release v1.0.0	✅	Release
Transaction success rate ≥ 90%	✅	96.7% (29 / 30) — §1 above
Proof generation ≤ 50 000 ms (NFR-2)	✅	744.6 ms avg — §1 above
Smart contracts deploy and function on Cardano testnet	✅	5 Preprod txs — §2 above
Real sandbox provider integration	✅	Wise transfer `2147582543` — §3 above
Midnight ZK circuit deployed	✅	4 SNARK proofs in blocks 15774→15784 — §4 above

8. Released artefacts

- `v1.0.0` **GitHub Release** — source archive: https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/releases/tag/v1.0.0
- **MIT LICENSE** — https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/blob/main/LICENSE
- **CHANGELOG** — https://github.com/Nucastio/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/blob/main/CHANGELOG.md
- **Specifications:** Project Initiation · SRS · TAD v1.1 (canonical)

9. Documentation link

Hosted developer documentation: https://nucastio.github.io/MidnightZK-Off-Ramp-SDK-ADA-Web2-Payments/

Rebuilt automatically on every change to `docs/` or `mkdocs.yml` via `.github/workflows/docs.yml`. The site covers integration, API reference, SDK reference, runnable examples, architecture, and the test/release evidence — sufficient for an external team to integrate the SDK without reading the source.